

Module 1 Homework Report

Foundational Exploratory Data Analysis & Data Dictionary

Course: MSBA 265

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Dataset: French Motor Third Party Liability Claims (freMTPL2freq)

Dataset Scale: 678,013 policy records × 12 variables

Purpose: Audit data quality, document business meaning, evaluate skewness and collinearity, interpret feature distributions, and establish reproducible outlier-cleaning logic.

Executive Result: The dataset contains no missing values, key operational ranges are valid, ClaimNb is a highly zero-inflated count target, no severe numerical multicollinearity is present, and Density requires special treatment because of its strong right tail.

1. Structural Audit and Data Quality

The structural audit used `df.info()` and `df.describe().T` to verify dataset scale, completeness, data types, and numerical bounds before downstream analytics.

1.1 Structural Findings

- **Dataset scale:** 678,013 individual policy records across 12 operational features.
- **Missing values:** every feature has 678,013 non-null entries (0% missing), so no missing-value imputation is required.
- **IDpol:** stored as float64, but it functions as a policy identifier rather than a continuous mathematical feature.
- **Categorical variables:** VehBrand, VehGas, Area, and Region are stored as object/string fields and require categorical encoding for later regression workflows.

1.2 Raw-Bound Verification

Feature	Observed Range/ Summary	Data-Quality Interpretation
DrivAge	18 to 100; median 44	Valid adult-policyholder range; no negative or underage sentinel errors detected.
Exposure	min ~ 0.002 ; mean ~ 0.53 ; max 1.0+	Positive risk-duration values; no negative exposure errors detected.
BonusMalus	50 to 230	Observed records remain within the documented regulatory/domain range.
ClaimNb	mean ~ 0.05 ; Q1=Q2=Q3=0	Extremely sparse count target with more than 95% zero observations.

1.3 Modeling Defense: Exposure Offset and Density/Area Redundancy

Exposure represents how long each insurance policy is observed. Policies with longer exposure have more opportunity to generate claims, so treat Exposure as a rate offset, typically using **$\log(\text{Exposure})$** , when modeling claim frequency. This allows the model to estimate claim rates rather than simply comparing raw claim counts.

Density is a continuous measure of population density, while **Area** is a categorical representation of density levels. Including both in the same linear regression model would introduce redundant information and may create multicollinearity, making coefficient estimates less stable and harder to interpret. Therefore, only one representation of this geographic-density information should be retained in a linear regression model.

2. Business Data Dictionary

The data dictionary connects technical field names to their operational meaning and intended analytical role. It is exported as **reports/data_dictionary.csv** in the project workflow.

Field	Business Meaning / Role
IDpol	Policy identifier; key field, not a continuous predictor.
ClaimNb	Number of claims; discrete count target.
Exposure	Policy risk exposure duration in years; rate-offset variable.
Area	Categorical geographic-density class (A-F).
VehPower	Vehicle power category.
VehAge	Vehicle age in years.
DrivAge	Driver age in years.
BonusMalus	Regulatory risk index.
VehBrand	Vehicle brand category.
VehGas	Vehicle fuel type.
Region	Geographic region category.
Density	Continuous population-density measure.

3. Skewness Diagnostics and ClaimNb Domain Override

The automated skewness audit flagged **ClaimNb** as extremely right skewed, with skewness approximately **5.60**. The automated transformation recommendation should not be followed blindly because ClaimNb is a discrete count outcome.

- Mathematical issue: more than 95% of ClaimNb values are exactly zero, and $\log(0)$ is undefined.
- $\log(x+1)$ issue: adding 1 changes the interpretation of the discrete count target and is not appropriate for the intended count-modeling framework.
- Domain interpretation: ClaimNb represents the number of claim events, not a continuous volume measure.
- Recommended action: keep ClaimNb in raw count units and use a count-modeling framework such as Poisson or Negative Binomial regression; Exposure should enter as a rate offset.

3.1 Key Observed Skewness Values

Feature	Skewness	Interpretation
ClaimNb	5.600	Extreme right skew; count-target override required.
Density	4.651	Strong right skew driven by high-density observations.
BonusMalus	1.729	Strong positive skew.
VehPower	1.171	Positive skew.
VehAge	1.148	Moderate positive skew.
DrivAge	0.436	Relatively mild skew.

4. Pearson Correlation Audit

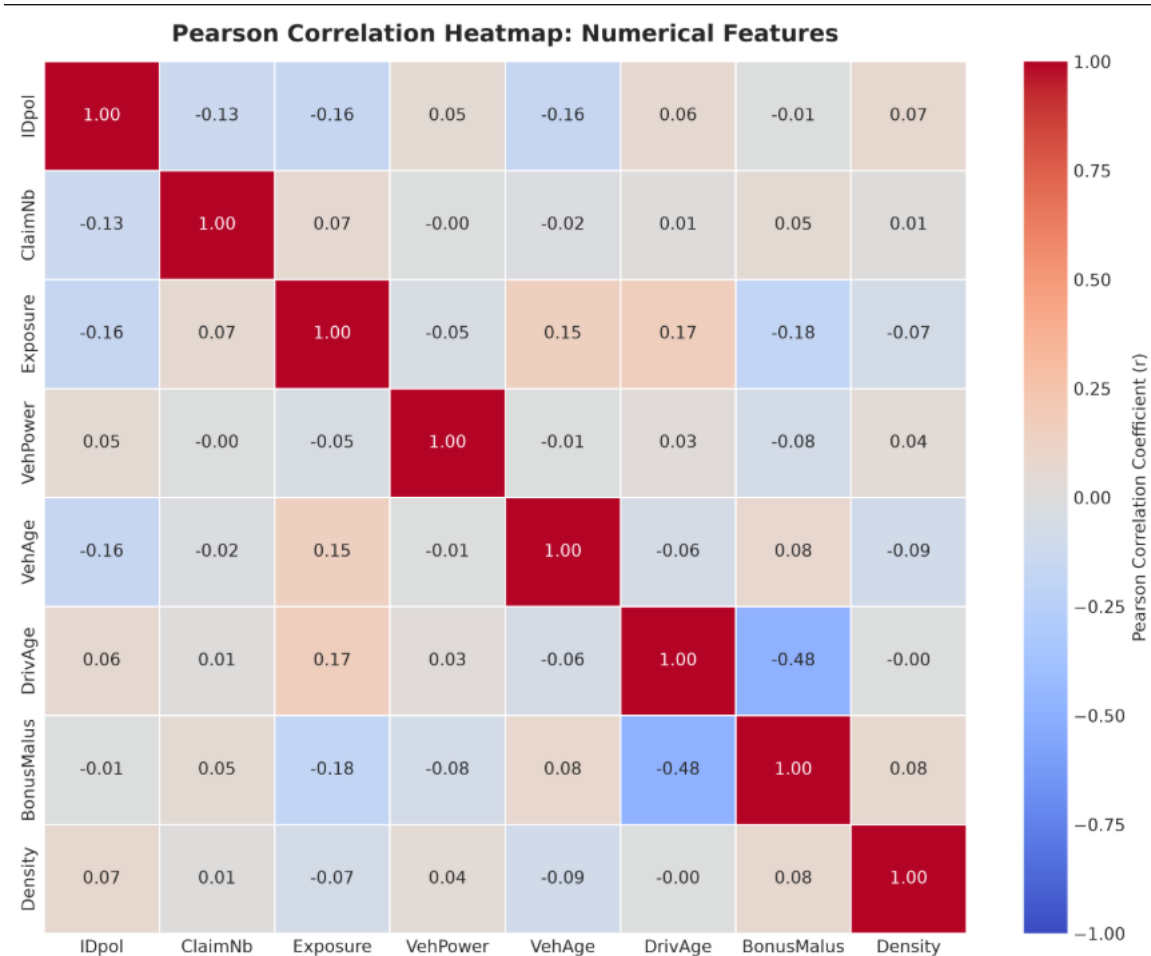


Figure 1. Pearson correlation heatmap exported by the notebook.

Required Interpretation

- **Distribution / Relationship Shape:** The heatmap shows generally weak linear relationships among most numerical variables. Excluding diagonal self-correlations, pairwise correlations range approximately from $r = -0.48$ to $r = +0.17$.
- **Key Metrics / Relationships:** The strongest observed numerical relationship is between DrivAge and BonusMalus, with a moderate negative correlation of approximately $r = -0.48$. no numerical pair reaches the severe multicollinearity threshold of $|r| \geq 0.85$.
- **Recommended Analytics Action:** No numerical predictor needs to be automatically removed based on this Pearson audit. Later modeling should still consider non-linear diagnostics and domain redundancy.

5. Distribution and Box-Plot Audit

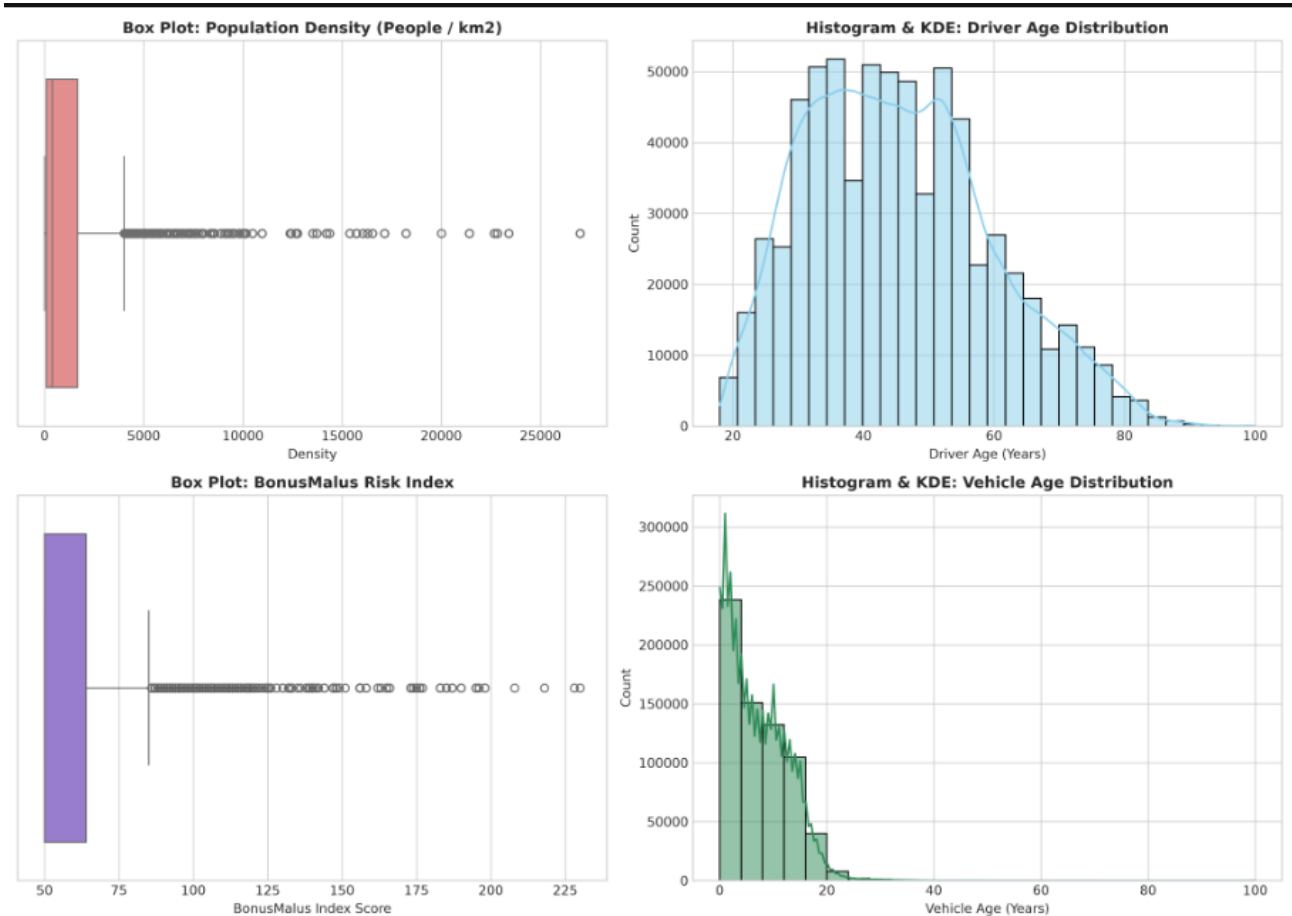


Figure 2. Feature distribution and box-plot audit for Density, DrivAge, BonusMalus, and VehAge.

5.1 Density

- **Distribution Shape:** strongly right-skewed; most policies occur in lower-density areas with a long metropolitan right tail.
- **Key Metrics / Whiskers:** Q1=92, median=393, Q3=1,658, IQR=1,566, upper Tukey fence=4,007, maximum=27,000.
- **Recommended Analytics Action:** do not automatically remove high-density records because they may be legitimate metropolitan observations; consider $\log(1 + \text{Density})$ for modeling.

5.2 Driver Age (DrivAge)

- **Distribution Shape:** roughly symmetric and bell-shaped around a median age of 44.
- **Key Metrics / Whiskers:** Q1=34, median=44, Q3=55, IQR=21, minimum=18, maximum=100.
- **Recommended Analytics Action:** retain the observed range; standard scaling may be used when required by the downstream model.

5.3 BonusMalus

- Distribution Shape: strongly right skewed, with many policyholders near the lower end and a smaller high-risk right tail.
- Key Metrics / Whiskers: Q1=50, median=50, Q3=64, IQR=14, upper Tukey fence=85, maximum=230.
- Recommended Analytics Action: retain legitimate high-risk records and consider a feature such as $\text{IsSurchargeDriver} = \text{BonusMalus} > 100$.

5.4 Vehicle Age (VehAge)

- Distribution Shape: moderately right skewed, with skewness approximately 1.15 and a concentration of younger vehicles.
- Key Metrics / Whiskers: Q1=2 years, median=6 years, Q3=11 years, with observed values extending to 100 years.
- Recommended Analytics Action: consider interpretable vehicle-age brackets or appropriate scaling/transformation; review extreme ages for business validity rather than automatically removing them.

6. Outlier Filtering and Reproducible Cleaning

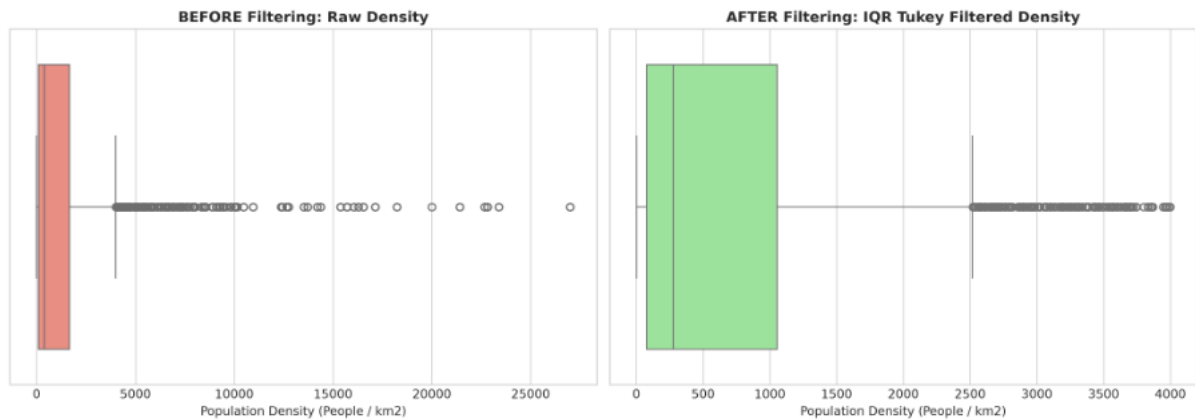


Figure 3. Density before/after outlier-filtering comparison from the notebook.

The reusable script `src/clean_outliers.py` applies the Tukey 1.5 x IQR rule to Density and writes the resulting dataset to `data/cleaned_business_data.csv`.

Metric	Result
Initial records	678,013
Density Tukey valid range	[-2,257, 4,007]
Records removed by Tukey filter	77,566 (11.44%)
Final cleaned records	600,447
Comparison: Z-score exclusions	15,209 (2.24%)

Density is highly skewed, so the mean and standard deviation are strongly influenced by the right tail. The Tukey method is therefore more robust for the production cleaning rule used in this assignment. Statistical outliers are not automatically data errors; the business meaning of high-density metropolitan observations must still be considered.

7. Executive Synthesis and Downstream Modeling Blueprint

- **Data quality:** the dataset is complete and key raw bounds were audited before modeling decisions were made.
- **Claim target:** ClaimNb should remain a discrete count target; Poisson is a natural baseline, and Negative Binomial is a potential alternative when dispersion diagnostics justify it.
- **Exposure:** use $\log(\text{Exposure})$ as the claim-frequency rate offset.
- **Density:** strong right skew supports transformation or robust treatment; Tukey filtering is encoded in the reproducible cleaning script required for this module.
- **DrivAge:** relatively symmetric distribution supports retaining the observed range and applying standard scaling only when required.
- **BonusMalus:** high values can represent legitimate surcharge-risk customers; retain business-valid observations and consider a surcharge indicator.
- **Area vs. Density:** treat these as potentially redundant representations of geographic density rather than automatically placing both into the same linear regression specification.
- **Reproducibility:** the project separates data procurement, notebook analysis, generated figures, data dictionary export, and production-style cleaning into reusable project artifacts.

Final Deliverables Referenced by This Report

data/download_data.py; notebooks/01edaanddatadictionary.ipynb; reports/data_dictionary.csv; reports/figures/correlation_heatmap.png; reports/figures/feature_distributions.png; src/clean_outliers.py; data/cleaned_business_data.csv; README.md; requirements.txt.

Conclusion: Module 1 establishes a reproducible, domain-informed EDA foundation for later predictive modeling. The analysis distinguishes mathematical flags from genuine business errors and translates each diagnostic into a specific modeling or preprocessing action.